DAV PUBLIC SCHOOL, SECTOR-4

DEPARTMENT OF MATHEMATICS

Subject: Class 11 Mathematics	Faculty: Ravi Shankar	Chapter: Intro to 3D Geometry

1. COORDINATE AXES & COORDINATE PLANES IN 3D SPACE

In two-dimensional geometry, a point is located using two perpendicular axes (X and Y). To locate a point in physical space, we extend this system by introducing a third axis, the **Z-axis**, perpendicular to both X and Y axes.

- **Origin (O):** The point where all three mutually perpendicular axes intersect. Its coordinates are **(0, 0, 0)**.
- **Coordinate Planes:** The three axes pair up to define three perpendicular planes:
  - **XY-Plane:** Contains the X and Y axes. Every point on this plane has **z = 0**.
  - **YZ-Plane:** Contains the Y and Z axes. Every point on this plane has **x = 0**.
  - **ZX-Plane:** Contains the Z and X axes. Every point on this plane has **y = 0**.
- **Octants:** The three coordinate planes divide space into **eight regions** called octants.

2. SIGNS OF COORDINATES IN THE EIGHT OCTANTS

The sign of the coordinates **(x, y, z)** depends on the octant in which the point lies:

Coordinate	I (X Y Z)	II (X' Y Z)	III (X' Y' Z)	IV (X Y' Z)	V (X Y Z')	VI (X' Y Z')	VII (X' Y' Z')	VIII (X Y' Z')
x	+	-	-	+	+	-	-	+
y	+	+	-	-	+	+	-	-
z	+	+	+	+	-	-	-	-

3. DISTANCE BETWEEN TWO POINTS IN 3D SPACE

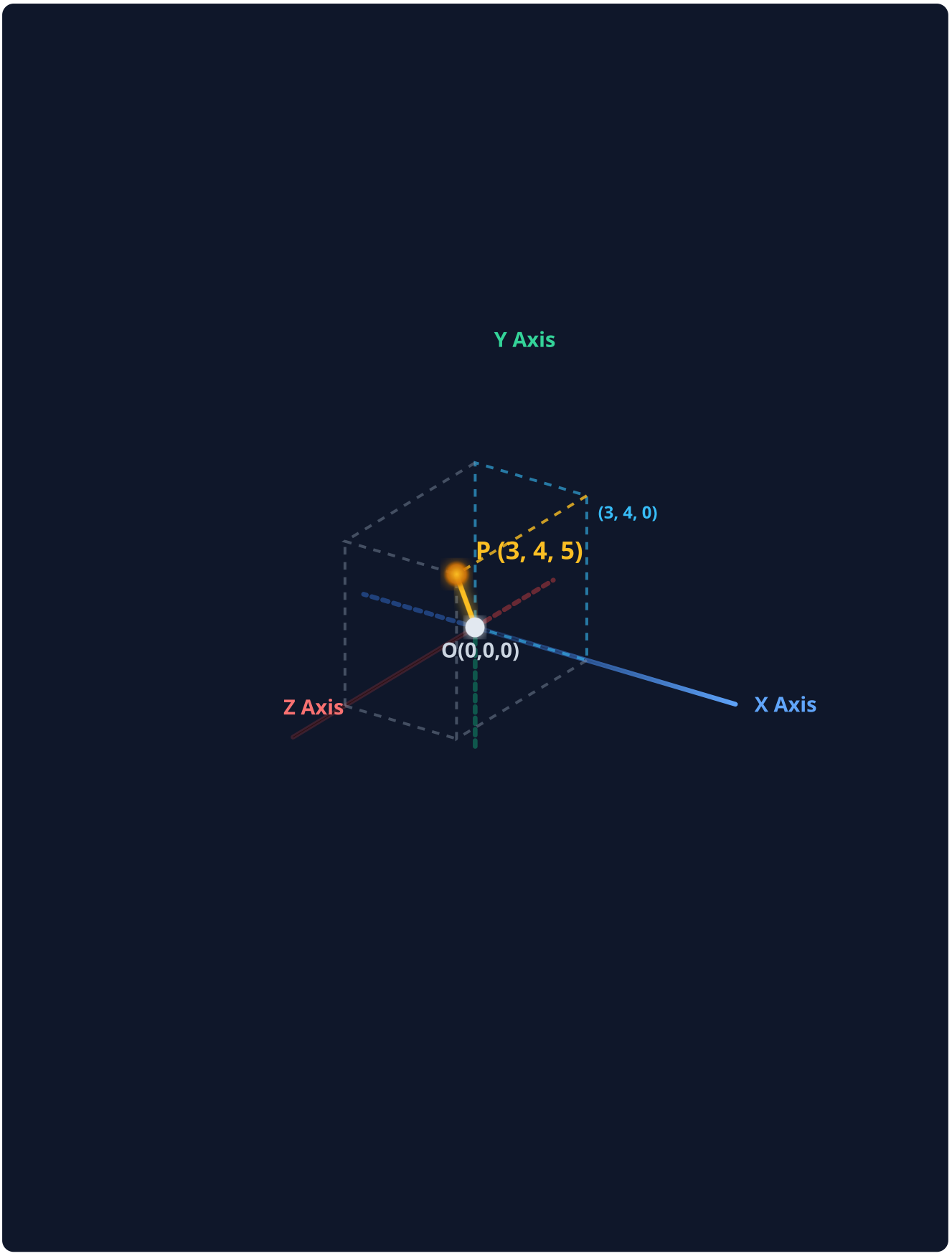
The distance between any two points $\mathbf{P(x_1, y_1, z_1)}$ and $\mathbf{Q(x_2, y_2, z_2)}$ is derived from the generalized Pythagorean Theorem:

Distance Formula:
$PQ = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]}$

Distance from Origin O(0,0,0):
$OP = \sqrt{[x^2 + y^2 + z^2]}$

4. INTERACTIVE VISUAL MODEL: POINT P(3, 4, 5) IN 3D SPACE

This interactive 3D rendering displays the coordinate axes, bounding box, projection vectors, and origin for point P(3, 4, 5). Click and drag inside the viewport to rotate space dynamically.



5. SECTION FORMULA (INTERNAL & EXTERNAL DIVISION)

Let **P(x_1, y_1, z_1)** and **Q(x_2, y_2, z_2)** be two points. The coordinates of point R dividing line segment PQ in ratio **m : n** are:

Internal Division (Ratio m : n):
R = [$(m x_2 + n x_1) / (m + n)$, $(m y_2 + n y_1) / (m + n)$, $(m z_2 + n z_1) / (m + n)$]

External Division (Ratio m : n):
R = [$(m x_2 - n x_1) / (m - n)$, $(m y_2 - n y_1) / (m - n)$, $(m z_2 - n z_1) / (m - n)$]

6. MIDPOINT AND CENTROID FORMULAS

- **Midpoint Formula:** For ratio 1 : 1, R = [$(x_1 + x_2)/2$, $(y_1 + y_2)/2$, $(z_1 + z_2)/2$]
- **Centroid of Triangle:** For vertices A, B, C: Centroid G = [$(x_1+x_2+x_3)/3$, $(y_1+y_2+y_3)/3$, $(z_1+z_2+z_3)/3$]

7. SOLVED PRACTICE EXAMPLES

Example 1: Distance Calculation
Question: Find the distance between P(-2, 3, 5) and Q(1, 2, 3).
Solution: PQ = $\sqrt{[(1 - (-2))^2 + (2 - 3)^2 + (3 - 5)^2]} = \sqrt{[3^2 + (-1)^2 + (-2)^2]} = \sqrt{[9 + 1 + 4]} =$ **$\sqrt{14}$ units**.

Example 2: Section Division
Question: Find coordinates of point R dividing A(1, -2, 3) and B(3, 4, -5) internally in ratio 2 : 3.
Solution: x = (2(3)+3(1))/5 = 9/5 | y = (2(4)+3(-2))/5 = 2/5 | z = (2(-5)+3(3))/5 = -1/5.
Point R: (9/5, 2/5, -1/5).

8. HOMEWORK ASSIGNMENT

1. Identify the octant containing the point **(-3, 1, -2)**.
2. Find the locus of points for which **y = 0** and **z = 0** simultaneously.
3. If origin O(0,0,0) is the centroid of triangle with vertices (2, a, 2), (b, 6, -5), and (4, 1, c), determine **a, b, and c**.
4. Prove that vertices A(0, 7, 10), B(-1, 6, 6), and C(-4, 9, 6) form an isosceles right-angled triangle.